When 2.0×10^{-4} mol of $XeF_6(s)$ is placed in 0.020 L of 0.30 M NaOH(*aq*), the following disproportionation reaction occurs:

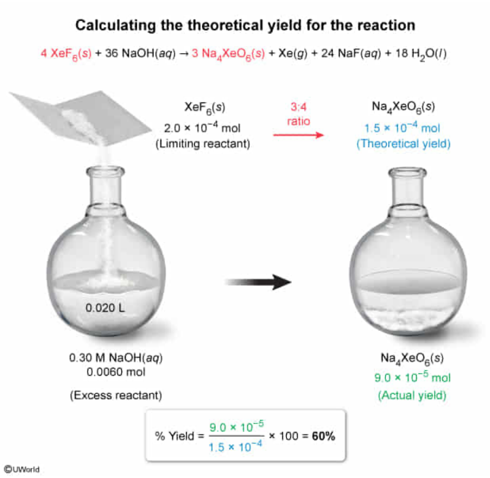
$$4\ XeF_6(s) + 36\ NaOH(aq) \rightarrow 3\ Na_4XeO_6(s) + Xe(g) + 24\ NaF(aq) + 18\ H_2O(l)$$

What is the percent yield of $Na_4XeO_6(s)$ for the reaction if 9.0×10^{-5} mol $Na_4XeO_6(s)$ is produced?

- A. 18% [18%]
- B. 33% [18%]
- C. 45% [18%]
- ✔ D. 60% [45%]

Correct
44% answered correctly

Explanation:



The **percent yield** for a reaction assesses the extent to which the starting materials are converted into products during a reaction. As such, the percent yield for a given product is defined as:

$$\%\ \text{Yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100$$

where the theoretical yield is the **maximum amount** of the product that can form if 100% of the limiting reactant is converted into products.

The given reaction includes two reactants (ie, XeF_6 and NaOH). As such, the limiting reactant must be determined by comparing the number of moles of each. The moles of $XeF_6(s)$ are given:

$$2.0 \times 10^{-4}\ \text{mol}\ XeF_6(s) = 0.00020\ \text{mol}\ XeF_6(s)$$

Multiplying the molar concentration of NaOH(*aq*) by the stated volume gives the moles of NaOH(aq):

$$0.020\ \cancel{\text{L NaOH(aq)}} \times \frac{0.30\ \text{mol NaOH(aq)}}{1\cancel{\text{L NaOH(aq)}}} = 0.0060\ \text{mol NaOH(aq)}$$

Based on the given reaction, every 4 molecules of $XeF_6(s)$ require 36 molecules of NaOH(*aq*) to react. Applying this molar ratio gives the number of moles required to convert all the $XeF_6(s)$.

$$0.00020\ \cancel{\text{mol}\ XeF_6(s)} \times \frac{36\ \text{mol NaOH(aq)}}{4\ \cancel{\text{mol}\ XeF_6(s)}} = 0.0018\ \text{mol NaOH(aq) needed}$$

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Because 0.0060 mol NaOH(*aq*) are present in the solution but only 0.0018 mol NaOH(*aq*) are needed, the **NaOH(*aq*) is in excess** and **XeF_6(s) is the limiting reactant**. Accordingly, if 100% of the moles of XeF_6(*s*) react to form Na_4XeO_6(*s*), the theoretical yield of Na_4XeO_6(*s*) in moles is found using the mole ratio from the reaction:

$$0.00020\ \cancel{\text{mol XeF}_6(s)} \times \frac{3\ \text{mol Na}_4\text{XeO}_6(s)}{4\ \cancel{\text{mol XeF}_6(s)}} = 0.00015\ \text{mol Na}_4\text{XeO}_6(s)\ \text{theoretical yield}$$

If 9.0×10^{-5} mol Na_4XeO_6(*s*) is produced (ie, the actual yield from the reaction), the percent yield is:

$$\%\ \text{Yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100 = \frac{0.00009\ \cancel{\text{mol Na}_4\text{XeO}_6(s)}}{0.00015\ \cancel{\text{mol Na}_4\text{XeO}_6(s)}} \times 100 = \frac{3 \times \cancel{(0.00003)}}{5 \times \cancel{(0.00003)}} \times 100 = 60\,\%$$

(Choice A) A value of 18% is obtained by calculating the theoretical yield on the incorrect assumption that NaOH(*aq*) is the limiting reactant.

$$0.0060\ \cancel{\text{mol NaOH(aq)}} \times \frac{3\ \text{mol Na}_4\text{XeO}_6(s)}{36\ \cancel{\text{mol NaOH(aq)}}} = 0.00050\ \text{mol Na}_4\text{XeO}_6(s)$$

$$\frac{0.00009\ \cancel{\text{mol Na}_4\text{XeO}_6(s)}}{0.00050\ \cancel{\text{mol Na}_4\text{XeO}_6(s)}} \times 100 = 18\,\%$$

(Choice B) A value of 33% is found if the $\frac{3\ \text{mol Na}_4\text{XeO}_6(s)}{4\ \text{mol XeF}_6(s)}$ mole ratio is mistakenly inverted as $\frac{4\ \text{mol XeF}_6(s)}{3\ \text{mol Na}_4\text{XeO}_6(s)}$ while calculating the theoretical yield. The compound units do not cancel properly.

$$0.00020\ \cancel{\text{mol}}\ \text{XeF}_6(s) \times \frac{4\ \text{mol XeF}_6(s)}{3\ \cancel{\text{mol}}\ \text{Na}_4\text{XeO}_6(s)} = 0.00027\ \text{mol}$$

$$\frac{0.000090\ \cancel{\text{mol}}}{0.00027\ \cancel{\text{mol}}} \times 100 = \frac{\cancel{3 \times (0.00003)}}{\cancel{9 \times (0.00003)}} \times 100 = \frac{1 \cancel{\times 3}}{3 \cancel{\times 3}} \times 100 = 33\,\%$$

(Choice C) A value of 45% is calculated if the actual yield is mistakenly divided by the moles of the XeF_6(*s*) reactant instead of dividing by the theoretical moles of Na_4XeO_6(*s*).

$$\frac{0.000090\ \cancel{\text{mol}}\ \text{Na}_4\text{XeO}_6(s)}{0.00020\ \cancel{\text{mol}}\ \text{XeF}_6(s)} \times 100 = \frac{9.0 \times 10^{-5}}{2.0 \times 10^{-4}} \times 100 = 45\,\%$$

Educational objective:
The percent yield for a reaction is calculated as the ratio of the actual yield (the amount obtained) to the theoretical yield (the amount possible) expressed as a percentage. This value assesses the extent to which the starting materials are converted into products during a reaction.

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